



Line-Following Robot

SWE Projects 2025

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01

Resources/ Materials

Know all your components !

Resources

- Line Following Robot
- Line Following Robot Resources

IR Sensors:

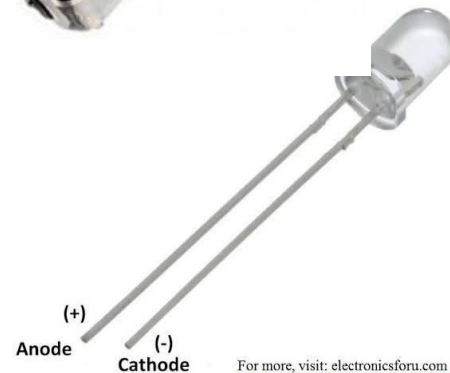
- [link](#)

Challenge Extension

- Complete a Figure-8 Course
- Navigate Intersections
- Follow Curved Paths
- Increase Maximum Speed
- Minimize Lap Time
- Complete a Maze
- Add Checkpoint Detection

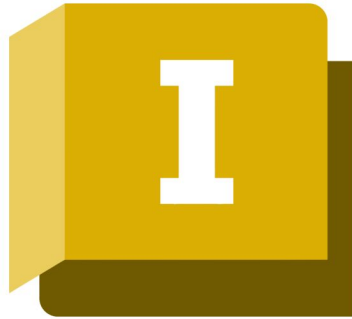
Know your Materials

- Elegoo Nano
- Normal USB cable with data transfer
- Breadboards & Wires
- Resistors (4.7 Kohms) for color sensor data clean
- Chassis/Frame - 3D Printed
- Power Supply (Battery)
- Black Electrical Tape for the line
- IR Sensors
- Motor Driver (L298N)
- DC Motors Dual Shafted
- Wheels
- Ball Roller



Bill of Materials:

https://docs.google.com/spreadsheets/d/1I8cKQe-k_1Rkc1F5U28vNx25RiU0Pge_27eqrAnHuNw/edit?usp=sharing



02

Design/CAD/ Mechanical

Plan out what you want to build !



Three Stages of Mechanical Prototyping



1. Design

Create sketches and plan the Line Following Robot features, layout, and interactions.



2. CAD

Develop a 3D model and integrated all electronic and mechanical components.



3. 3D Print

Fabricate and test the prototype, refining the design through multiple iterations.

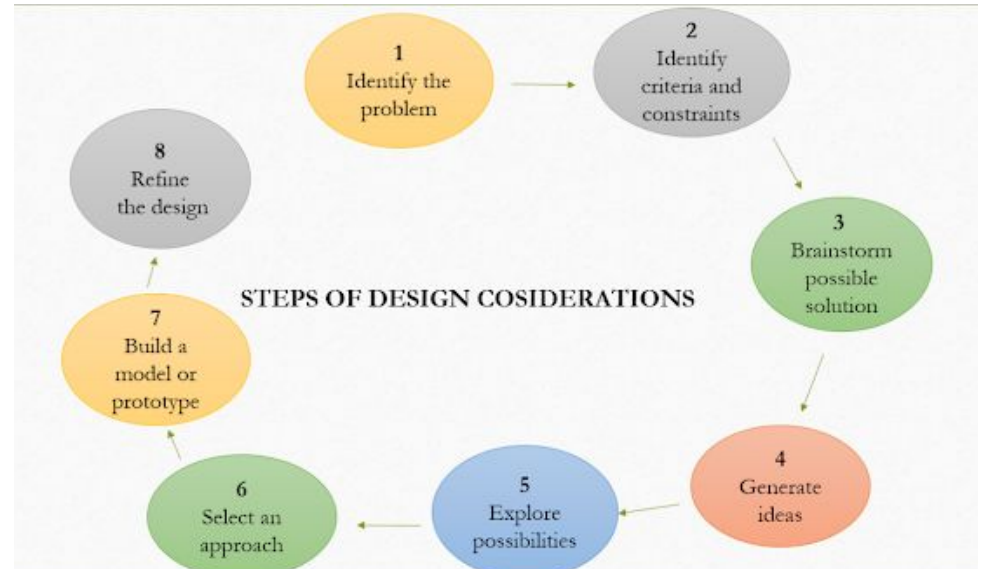
Engineering Design Constraints & Considerations

Design Considerations

Where will each component fit?
 How do we route wires cleanly?
 Will the design be easy to assemble?
 Can the structure support the hardware?
 How do we make the design compact and functional?

Mechanical Planning

Measuring motors, sensors, and batteries
 Planning mounting locations
 Designing around real component dimensions
 Balancing aesthetics and functionality



Students learned that engineering design starts long before fabrication.

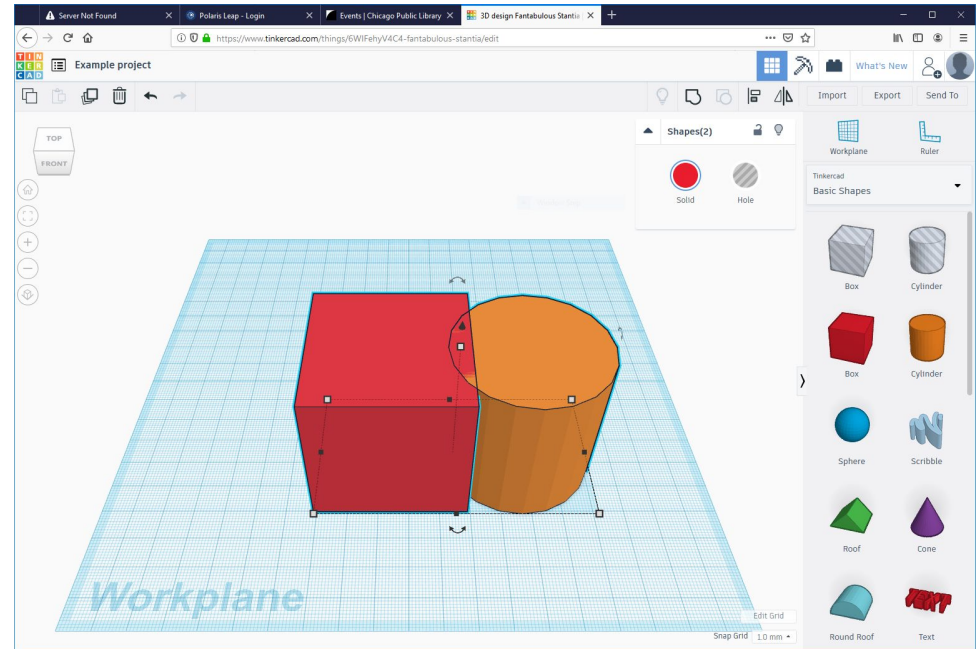
CAD - Finalize Design/3D Printing

CAD Development

- Creating initial sketches and part layouts
- Iterating designs based on component fit
- Learning tolerances and clearances
- Preparing models for fabrication

3D Printing & Prototyping Planning

- Exporting STL files for printing
- Testing printed parts for fit/function
- Adjusting designs after failed prints
- Understanding real-world manufacturing constraints
- Turning ideas into physical prototypes



Members gained hands-on experience with the full prototyping workflow.



Inventor or TinkerCAD

Inventor - AutoDesk



- Similar to SolidWorks
- Industry-standard CAD software
- Great introduction to mechanical design workflows
- Helps build technical portfolio projects
- Best for students wanting deeper CAD experience

Download:

<https://www.autodesk.com/education/edu-software/overview#INVPROSA>

TinkerCAD - AutoDesk



- Beginner-friendly CAD platform
- Easy entry point for first-time designers
- Quick learning curve for workshops/projects
- Great for Arduino + 3D printing integration
- Helps students gain confidence before advanced CAD

3D Design TinkerCAD:

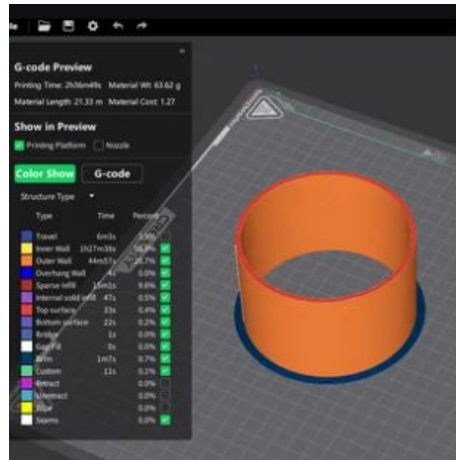
<https://www.tinkercad.com/3d-design>

These softwares chosen to make CAD more accessible for students of all experience levels.

3D Printing Slicer Software (CAD → 3D Printer)

CAD are in .STL Files
3D Printers can only read G-code

Slicer software converts .STL files to G-code instructions



Basic Slicer Workflow

- Import .STL file into slicer software
- Position model flat on the print bed
- Optimize orientation for stronger/more successful prints
- Scale or resize if necessary
- Add supports for overhangs or floating sections
- Check layer preview before printing
- Export final G-code to printer

Good Printing Practices

- Minimize unnecessary supports
- Avoid tall unstable orientations
- Keep large flat surfaces on the print bed
- Watch for unsupported overhangs
- Verify print time and material usage before printing



Ender3 Printer → Creality Slicer

Different printers may use different slicer software.

SWE projects used the Ender 3 printer with Creality Slicer.

Download Slicer Software "Creality Print":

<https://www.creality.com/download/ender-3-3d-printer>



03

Electrical/ Schematics

Wire it up, No wires = No programming...

Black Box Diagrams - Visualization

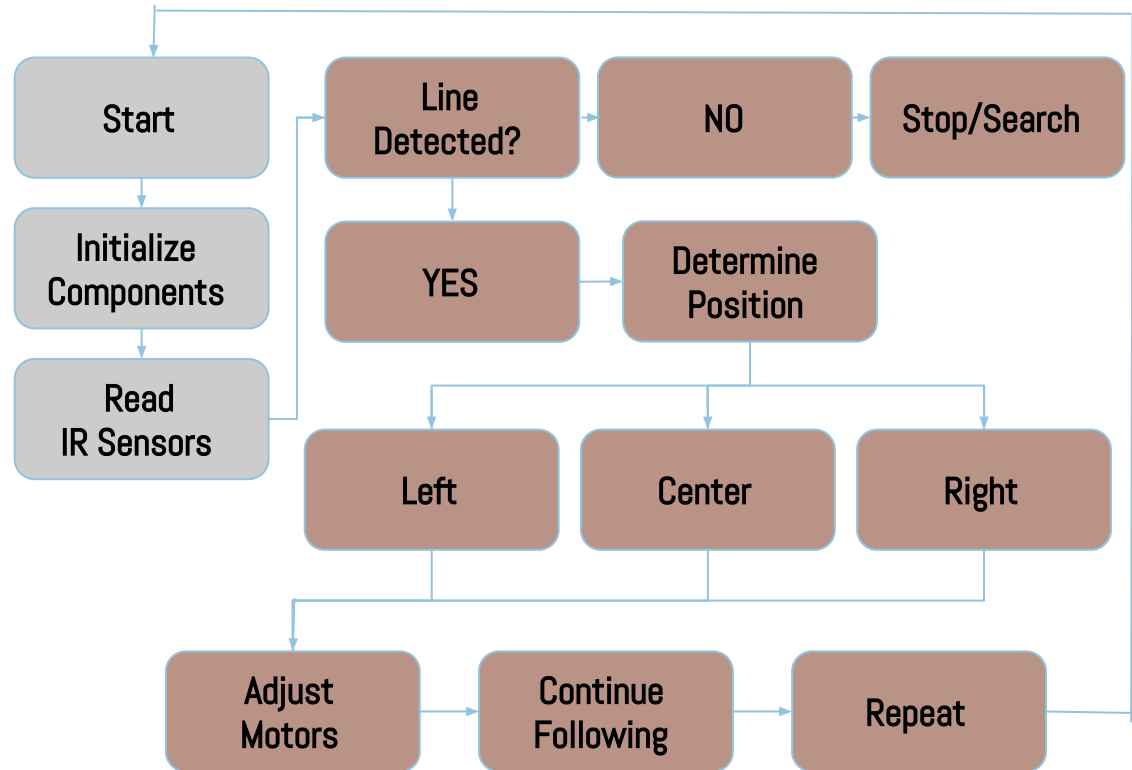
Black Box Diagram



Explains what the system does by showing inputs, outputs, and the system boundary.

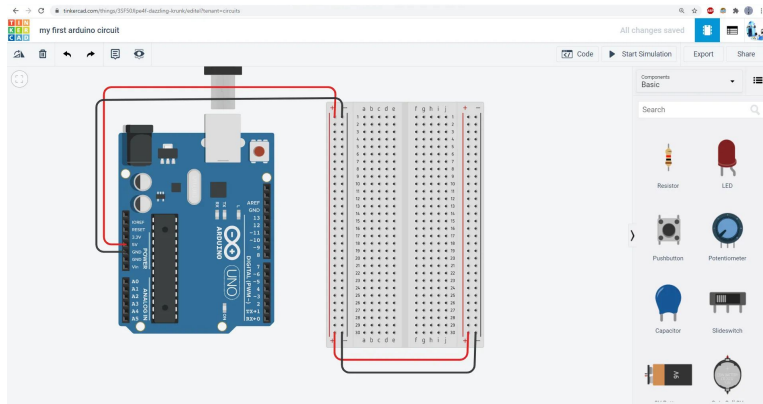
Flow Diagram/Flow Chart

Illustrates the sequence of operations and decision-making within the program. It helps visualize how the robot processes sensor inputs to detect and follow a line autonomously.



Breadboard

TinkerCAD



Resources

Reading:

<https://www.build-electronic-circuits.com/breadboard/>

Videos:

https://youtu.be/xs8No_rg40U?si=nRS4d0bBDzABbNn

Intro Slides: https://docs.google.com/presentation/d/1yD5nXMJWPMtZPHhtXyCd_YX-FAF2YCUzgcZnmeqP8_c/edit?usp=sharing

[illegible]

Components (Motors)

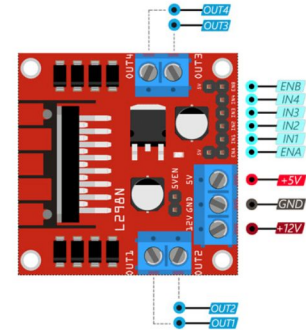
DC Motor



Provides the driving force for the robot by converting electrical energy into rotational motion. Two DC motors independently control the left and right wheels, allowing the robot to move forward, turn, and follow a line.

Motor Driver (L298N)

Controls the speed and direction of the DC motors using signals from the Arduino. It supplies sufficient power to the motors while enabling precise movement and steering for line following.



Components (Input)



Infrared LED (IR Emitter)

Emits infrared light that is reflected by nearby surfaces. It works together with the IR receiver to detect changes in surface color, enabling the robot to distinguish between the line and the surrounding area.

Infrared Receiver

Detects infrared light reflected from the surface and converts it into an electrical signal. The Arduino uses this feedback to determine the robot's position relative to the line and adjust its movement accordingly.



Components (Output)

Passive Buzzer



Buzzer (Passive)

Produces customizable tones using different frequencies controlled by the Arduino. It can provide audible feedback for startup, successful calibration, line detection, or completion of the robot's course.

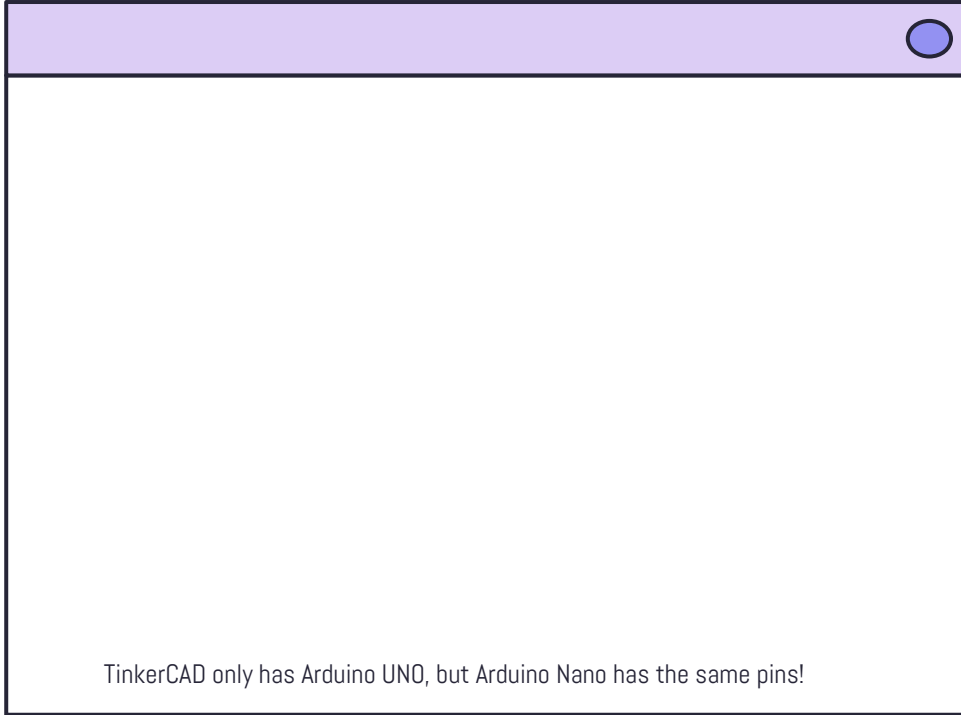
Active Buzzer



Buzzer (Active)

Generates a simple alert sound when powered. It is commonly used to indicate power-on, errors, obstacle detection, or changes in the robot's operating state.

Sample Circuit - Customize your Own !!



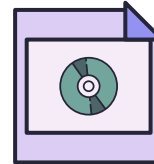
Nano

main microcontroller,
reading sensor inputs
and controlling outputs



Power

Distributes 5V and
GND from the Arduino
to the breadboard,



Jumper Wires

Connect the Arduino,
breadboard, and
electronic components



Resistors

Limit current to
protect components
and ensure safety



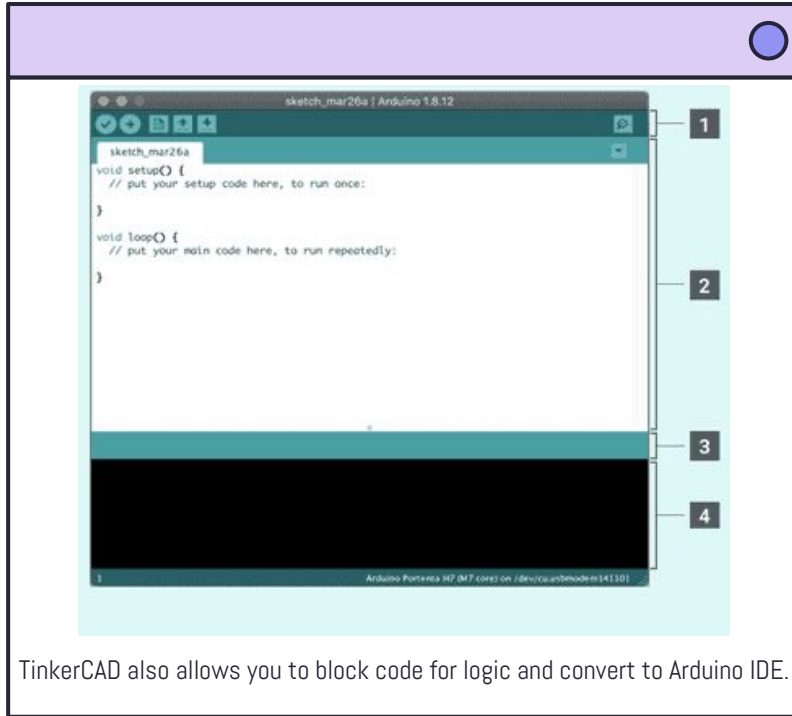
04

Programming

Watch your build come to life !!



Arduino IDE (C++) -

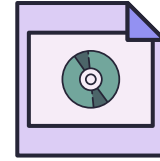


TinkerCAD also allows you to block code for logic and convert to Arduino IDE.



Download

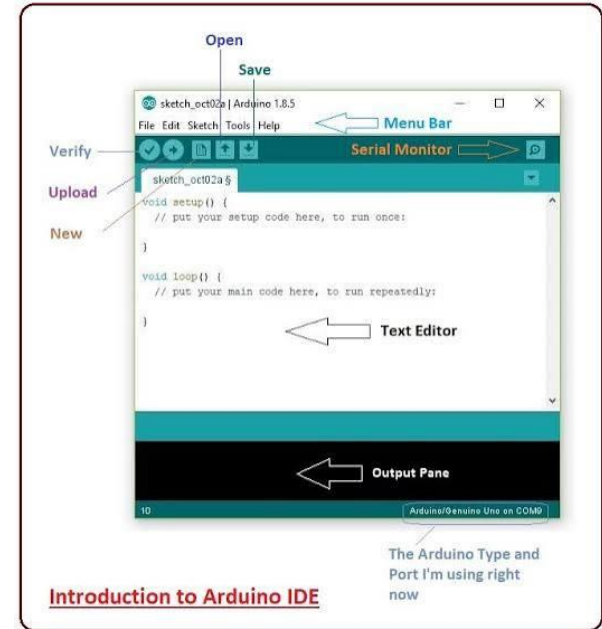
<https://support.arduino.cc/hc/en-us/articles/360019833020-Download-and-install-Arduino-IDE>



Arduino IDE Function Library

<https://docs.arduino.cc/language-reference/>

Arduino IDE - Must Know



Arduino IDE Intro: <https://docs.arduino.cc/software/ide-v2/tutorials/getting-started-ide-v2/>



Arduino IDE - Parts

The screenshot shows the Arduino IDE window titled "Blink | Arduino 1.0". The code editor contains the following text:

```
/*
 * Blink
 * Turns on an LED on for one second, then off for one second, repeatedly.
 *
 * This example code is in the public domain.
 */

void setup() {
  // initialize the digital pin as an output.
  // Pin 13 has an LED connected on most Arduino boards:
  pinMode(13, OUTPUT);
}

void loop() {
  digitalWrite(13, HIGH); // set the LED on
  delay(1000);            // wait for a second
  digitalWrite(13, LOW);  // set the LED off
  delay(1000);            // wait for a second
}
```

At the bottom of the window, a status bar indicates "1" and "Arduino Uno on /dev/tty.usbmodem621".

void setup ()

Setup Function

Runs once when the Arduino powers on. Used to initialize components such as sensors, servos, buzzers, and the LCD display before the Desk Buddy begins operating.

void loop ()

Main Program Loop

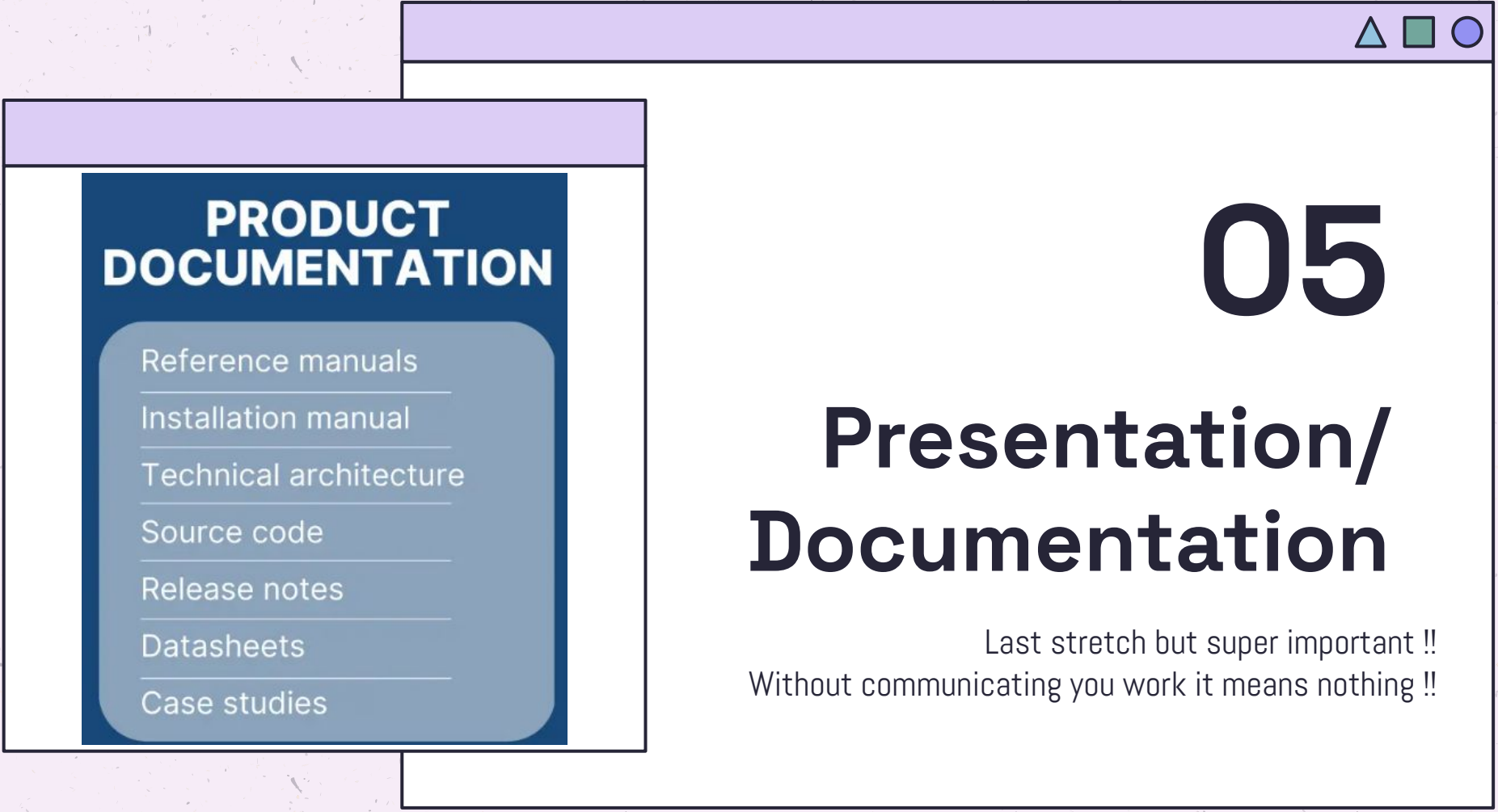
Runs continuously after `setup()`. It repeatedly checks sensor inputs and updates the Desk Buddy's actions, displays, sounds, and movements.

Variables

Store information used throughout the program, such as temperature readings, motion detection status, servo positions, and user settings.

Functions

Reusable blocks of code that perform specific tasks. Functions help organize the program and simplify actions like displaying messages, playing sounds, or moving servos.



PRODUCT DOCUMENTATION

Reference manuals

Installation manual

Technical architecture

Source code

Release notes

Datasheets

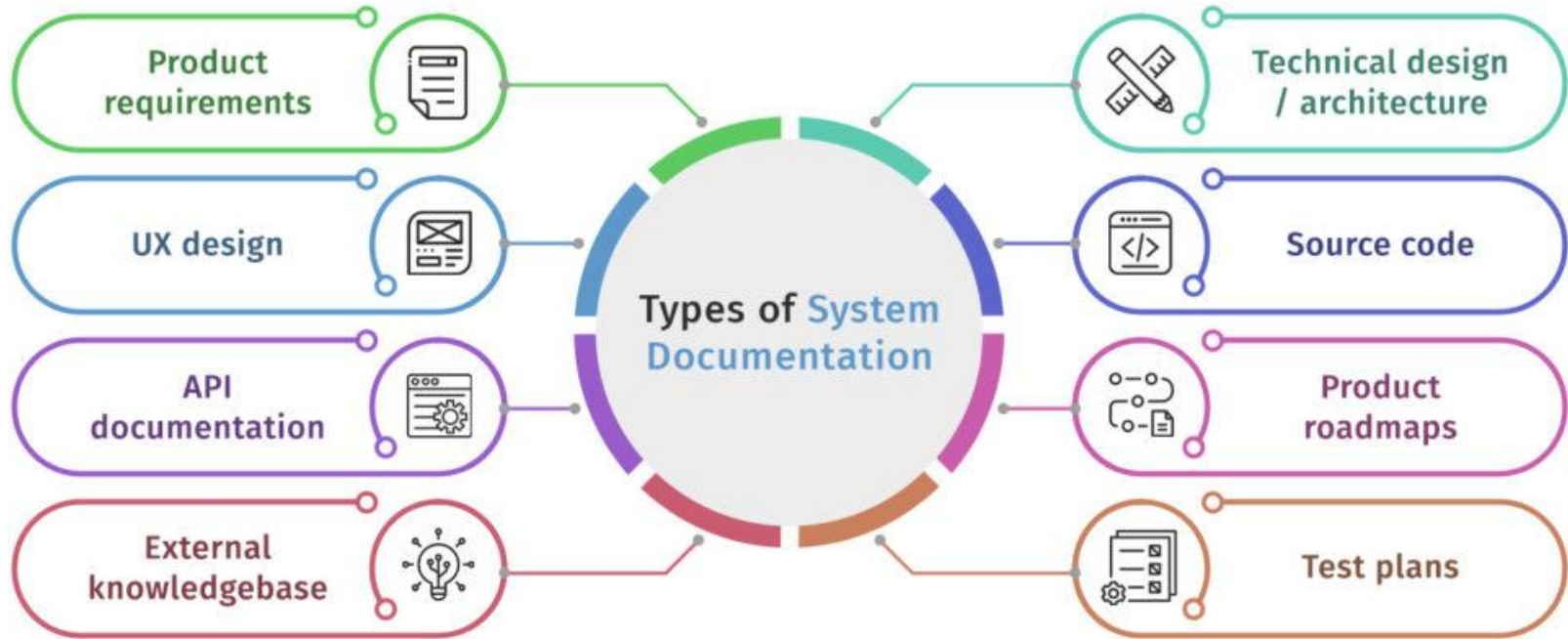
Case studies

05

Presentation/ Documentation

Last stretch but super important !!
Without communicating your work it means nothing !!

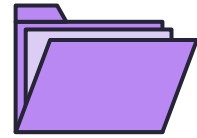
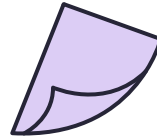
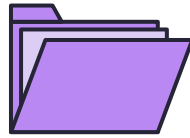
Product Based Documentation !!



<https://technicalwriterhq.com/documentation/product-documentation/>



Presentation - 4 parts



Design/CAD

CAD Files

Schematics

Circuit Schematic
File

Program w/ Comments

Arduino IDE Code
w/ Comments

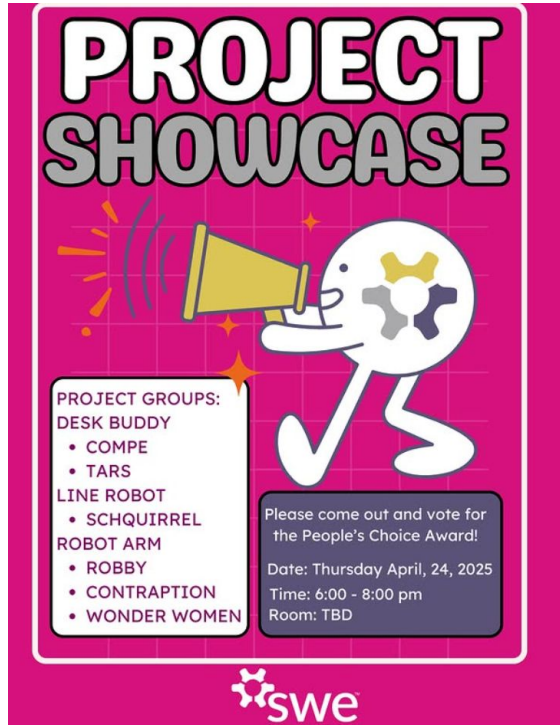
Pictures of final Product

Pictures/Videos

Presentation

Presentation Template:

SWE Project Showcase (Demo) - Awards !!



Please have your
Project + Slides ready for our
Inaugural SWE Project Showcase !!

Thursday, April 24th
6:00 - 8:00pm
Arch 310

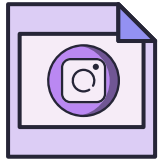


Thanks!

Do you have any questions?



Please Contact Me !!
Project.swe.uh@gmail.com
@Tako_124 on Discord



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Troubleshooting Help - Common



won't follow the line

Verify IR sensor wiring

Adjust sensor height

Calibrate sensor
sensitivity



Motors not moving

Check L298N power supply

Verify motor wiring

Confirm ENA/ENB jumper
settings



Robot turns the wrong way

Swap left/right motor wires

Reverse motor direction in code

Verify IR sensor order

Further Learning

- D

Keep Learning !!

Future Improvements

- PID Line Following
- Adjustable Speed Control
- Obstacle Avoidance
- Bluetooth Control
- OLED/LCD Status Display
- Maze Solving
- Rechargeable Battery Pack
- Autonomous Mapping

